Mobile-Based Virtual Physics Laboratory with AI Assistance and Interactive Learning for Rural Students

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Physics education is essential for developing scientific thinking and problem-solving skills among school students. However, many learners struggle to understand physics concepts due to the abstract nature of theoretical explanations and the lack of practical experimentation opportunities. This issue is particularly significant in rural schools where laboratory facilities and learning resources are limited. As a result, students often learn physics mainly through textbooks and lectures, which reduces engagement and makes it difficult to connect theoretical formulas with real-world observations.

Although several digital learning platforms and simulations have been introduced to support science education, most existing systems focus only on single learning approaches such as simulations, quizzes, or gamified content. These tools often lack integration between interactive activities, measurable experimental results, and step-by-step explanations of physics calculations. Consequently, students may interact with simulations or games without fully understanding the mathematical relationships behind the observed outcomes.

To address this limitation, this research proposes the development of an Interactive Physics Experimentation and Learning Module designed for Grade 6–11 students, particularly in rural educational environments. The proposed system integrates physics-based games, real-time measurement generation, step-by-step calculation explanations, and a virtual practical laboratory environment within a mobile application. In this module, students can interact with physics activities such as motion-based simulations where their actions generate measurable data (e.g., distance travelled or motion). The system then explains the relevant physics formulas and calculations to help learners understand how theoretical concepts are applied in practice. Additionally, the virtual laboratory allows students to conduct digital experiments using simulated laboratory tools.

The system will be developed using mobile application technologies, interactive simulation frameworks, and physics-based algorithms for measurement calculation and visualization. The expected outcomes include improved conceptual understanding of physics, increased student engagement, and greater accessibility to experimental learning for students who lack laboratory facilities.

From a social perspective, the system supports inclusive and equitable quality education, particularly benefiting rural learners. Commercially, the platform has potential for adoption by schools and educational institutions as an affordable digital learning tool.

Keywords: Physics Education, Game-Based Learning, Virtual Laboratory, Interactive Simulations, Mobile Learning.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
CV	Computer Vision
YOLO	You Only Look Once
YOLOv8	You Only Look Once Version 8(Object Detection Model)
UI	User Interface
UX	User Experience
DSR	Design Science Research
API	Application Programming Interface
DB	Database
SDG	Sustainable Development Goal
ICT	Information and Communication Technology
STEM	Science, Technology, Engineering, and Mathematics
FPS	Frames Per Second

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1 INTRODUCTION

1.1 Background and Context

Physics education continues to face significant challenges due to students' low engagement, poor conceptual understanding, and limited access to practical laboratory experiences. Many learners perceive physics as difficult, abstract, and disconnected from real-life applications, which negatively impacts motivation and academic performance [1] This issue affects secondary school students globally, particularly in developing countries, where access to laboratory facilities and modern learning tools is limited, making it difficult to achieve effective STEM education outcomes [2].For instance, international assessments such as PISA highlight low student performance in science subjects, emphasizing the urgency for improved teaching methodologies [1].

Existing solutions such as virtual laboratories and simulation tools have been introduced to address these challenges. Virtual labs enable students to perform experiments remotely and support observation, measurement, and analysis processes [3]. Similarly, interactive simulations like PhET have demonstrated improvements in student understanding, engagement, and academic performance [4] [5]. However, most of these systems focus mainly on visualization and conceptual learning rather than integrating real-time measurement generation, calculation processes, and structured experiment evaluation. Additionally, many systems operate as standalone tools and do not provide a comprehensive learning experience that combines gameplay, experimentation, and automated feedback in a single platform [3] [5].

To overcome these limitations, this research proposes an Interactive Physics Experimentation and Learning Module that integrates game-based learning with virtual experiments. The system generates real-time measurements (e.g., distance, time, velocity) from student interactions, applies physics formulas, and provides step-by-step calculation explanations. It also evaluates student performance based on apparatus selection, safety practices, procedures, and result accuracy. The system is developed using Unity 2D, OpenCV, YOLOv8, Flutter, Python, and Firebase. This solution supports Sustainable Development Goal 4 (Quality Education) by improving access to interactive and practical learning. Furthermore, it aligns with the AI-driven Educational Technology research cluster, promoting innovative, scalable, and student-centered digital learning solutions.

1.2 Literature Review

Recent research in physics education shows a growing shift from traditional lecture-based teaching toward interactive digital learning approaches such as simulations, virtual laboratories, and game-based learning. A major research approach in the literature is the use of interactive simulations to improve students' conceptual understanding, motivation, and academic performance. Studies based on PhET simulations report that simulation-based learning helps students visualize abstract concepts, engage actively with content, and achieve better learning outcomes than conventional teaching methods [1] [6] [7] [8]. These studies also show that simulation-supported learning is especially useful in secondary school physics, where many learners struggle with difficult topics and low motivation [1] [8].

Another important research approach is the use of computer simulations as substitutes or supplements for laboratory activities. Foundational studies found that well-designed simulations can support conceptual understanding and even help students perform better than those using only traditional equipment [2] [3]. Similarly, virtual laboratory research highlights that digital labs can support observation, measurement, data collection, analysis, and conclusion drawing when real laboratory access is limited [9]. These approaches are valuable for physics education because they reduce dependence on physical resources and provide more accessible practical learning opportunities.

A third stream of literature focuses on game-based learning and gamification. Research shows that educational games can make science learning more relevant, engaging, and enjoyable, while also improving content understanding [4]. Studies on gamification further suggest that game elements such as points, badges, challenges, and progression can increase learner motivation and participation [5] [10]. In addition, recent work has explored PhET not only as a simulation tool but also as a game-based learning environment, reporting improvements in both understanding and learner interest [6]. These findings are relevant to your component because they support the use of interactive and game-like environments in physics learning.

In terms of existing products and services, the reviewed studies mainly discuss tools such as PhET interactive simulations**,** virtual laboratory systems**,** and educational physics games like *Physicus* [2] [4] [9] [6]. PhET is the most frequently highlighted product because it is interactive, visually engaging, and widely used in school science education [1] [2] [6] [7] [8] Virtual lab systems are presented as useful alternatives when schools lack sufficient laboratory facilities [9], while physics games are discussed as tools that increase enjoyment and help students learn specific concepts through exploration and problem-solving [4]

However, the literature also reveals several limitations in existing systems. First, most simulation-based tools mainly support conceptual visualization and basic interaction, but they do not generate real-time measurable outputs such as distance, time, or velocity from student actions [1] [6] [8]. Second, many systems improve engagement but do not connect student interaction to automatic physics formula application and step-by-step calculation explanations [4] [10] Third, virtual labs often support observation and measurement, yet they rarely provide structured evaluation of student performance based on apparatus selection, safety tools, procedure steps, and final result accuracy [9]. Fourth, many existing tools function as separate

systems—simulations, games, tutoring environments, and virtual labs are usually designed independently rather than integrated into one learning platform [3] [9] [10] Finally, reviews of gamification indicate that although game elements can increase motivation, there is still limited evidence on how to design them effectively for long-term educational value [5] [10].

Overall, the literature confirms that simulations, virtual labs, and game-based learning can enhance physics education, but current systems remain fragmented. There is still limited research on an integrated mobile learning module that combines game-based interaction, real-time measurement generation, automatic calculation support, and virtual experiment evaluation within one platform. This gap provides the basis for the proposed Interactive Physics Experimentation and Learning Module**.**

1.3 Research Gap

Although existing studies confirm that simulations, virtual laboratories, and game-based learning can improve physics education, an important gap still remains in how these approaches are designed and integrated. Current systems mainly focus on concept visualization, motivation, or isolated virtual practice, but they do not provide a unified learning process that connects student interaction, measurable outputs, calculation reasoning, and experiment assessment in one platform [1] [3] [9] [6]. For example, PhET and similar interactive simulations are effective for improving understanding and engagement, yet they are mainly designed for exploration and conceptual learning rather than for generating real-time physics measurements from gameplay and converting them into structured calculation explanations [1] [2] [6] [8].

Likewise, virtual laboratory systems support observation, measurement, and data analysis, but they often function only as digital substitutes for experiments and do not evaluate students in a detailed manner based on apparatus selection, safety tools, procedural steps, and result accuracy [9]. Game-based learning studies show that educational games can enhance motivation, enjoyment, and content learning, but most of these systems stop at engagement and do not connect gameplay to physics formula application or measurable academic outputs such as distance, time, and velocity [4] [5] [10]. In addition, many existing approaches are implemented as separate tools—simulations for visualization, games for engagement, and virtual labs for experimentation—without offering an integrated mobile-based environment that combines all these learning functions [3] [9] [10].

Therefore, the key research gap is the absence of a single intelligent physics learning module that integrates game-based interaction, real-time measurement generation, automatic step-by-step calculation support, and structured virtual experiment evaluation. This makes current approaches insufficient for supporting both conceptual learning and practical skill development in a complete and continuous way [4] [9] [6]. The novelty of this research lies in proposing an Interactive Physics Experimentation and Learning Module that combines these missing elements in one mobile platform using Unity 2D, OpenCV, YOLOv8, Flutter, Python, and Firebase. This integrated design is creative because it moves beyond simple simulations or gamification and introduces a system where student actions directly produce measurable values, trigger formula-based reasoning, and support practical experiment assessment. Hence, this study contributes a more comprehensive and innovative solution to the current limitations in digital physics education.

2. OBJECTIVE

2.1 Main Objective

To develop and evaluate an interactive physics learning system that integrates game-based measurement, automated calculation, and virtual experimentation to improve conceptual understanding and practical skills of Grade 6–11 students.

2.2 Specific Objectives

- To capture real-time physics measurements (e.g., distance, time, velocity) from student interactions using object detection techniques.
- To apply physics formulas automatically and generate step-by-step calculation explanations based on measured values.
- To design and implement virtual laboratory experiments aligned with school physics curriculum using interactive digital tools.
- To evaluate student experiment performance based on apparatus selection, safety practices, procedural steps, and accuracy of results.
- To develop a mobile-based application that integrates games, calculations, and experiments into a user-friendly learning platform.
- To assess the effectiveness of the system in improving student engagement and understanding of physics concepts through user testing and performance analysis.

3. METHODOLOGY

Research Approach

This research follows a Design Science Research (DSR) approach combined with experimental evaluation. The Design Science approach is used to design and develop an innovative interactive physics learning system that integrates games, measurement, and virtual experiments. The experimental approach is applied to evaluate the effectiveness of the system in improving students' understanding, engagement, and performance in physics learning. This combination ensures both system development and validation in a real educational context.

Technical Methods and Algorithms

The system uses several technical methods to achieve its functionality. Interactive physics games and virtual experiments are developed using simulation techniques, where student actions generate measurable values such as distance, time, and velocity. Computer vision methods using OpenCV and YOLOv8 are applied for object detection and motion tracking to extract these measurements from simulation environments. A rule-based calculation engine is implemented to apply physics formulas and generate step-by-step explanations. Additionally, a performance evaluation algorithm is designed to assess experiments based on criteria such as apparatus selection, procedural correctness, safety considerations, and result accuracy.

Tools and Platforms with Justification

The system is developed using a combination of modern tools and platforms. Unity 2D is used to create interactive physics games and virtual experiments due to its strong support for real-time simulation and user interaction. OpenCV and YOLOv8 are used for object detection and measurement extraction because of their accuracy and efficiency in handling motion-related data. Flutter is selected for mobile application development as it enables cross-platform compatibility and a user-friendly interface**.** Python is used for backend processing and implementing calculation and evaluation logic due to its flexibility and strong library support. Firebase is used for cloud storage and real-time data management, ensuring scalability and accessibility.

Data Collection and Ethical Considerations

The system collects data related to student interactions, including generated measurements, calculation results, and experiment performance. This data is used to evaluate learning outcomes and system effectiveness. Ethical considerations are maintained by ensuring that no sensitive personal information is collected, and all data is used strictly for educational and research purposes. User consent is obtained where necessary, and data privacy and confidentiality are ensured through secure storage mechanisms.

Validation Strategy and Performance Metrics

The system is validated through user testing with students in the target group (Grade 6–11). The evaluation focuses on both system performance and learning outcomes. Key performance metrics include accuracy of measurement extraction**,** correctness of generated calculations, and reliability of experiment evaluation. Educational effectiveness is measured through improvements in student understanding, engagement levels, and task completion rates. User feedback is also collected to assess usability and overall learning experience.

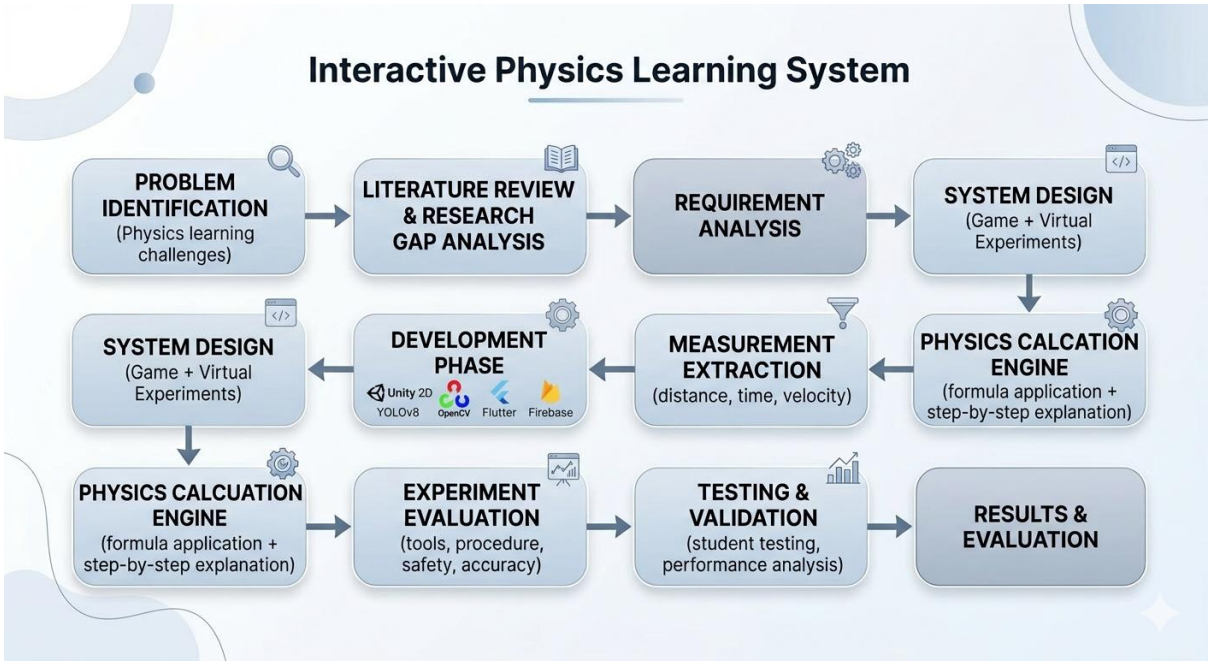


Figure 1 Research Methodology Diagram

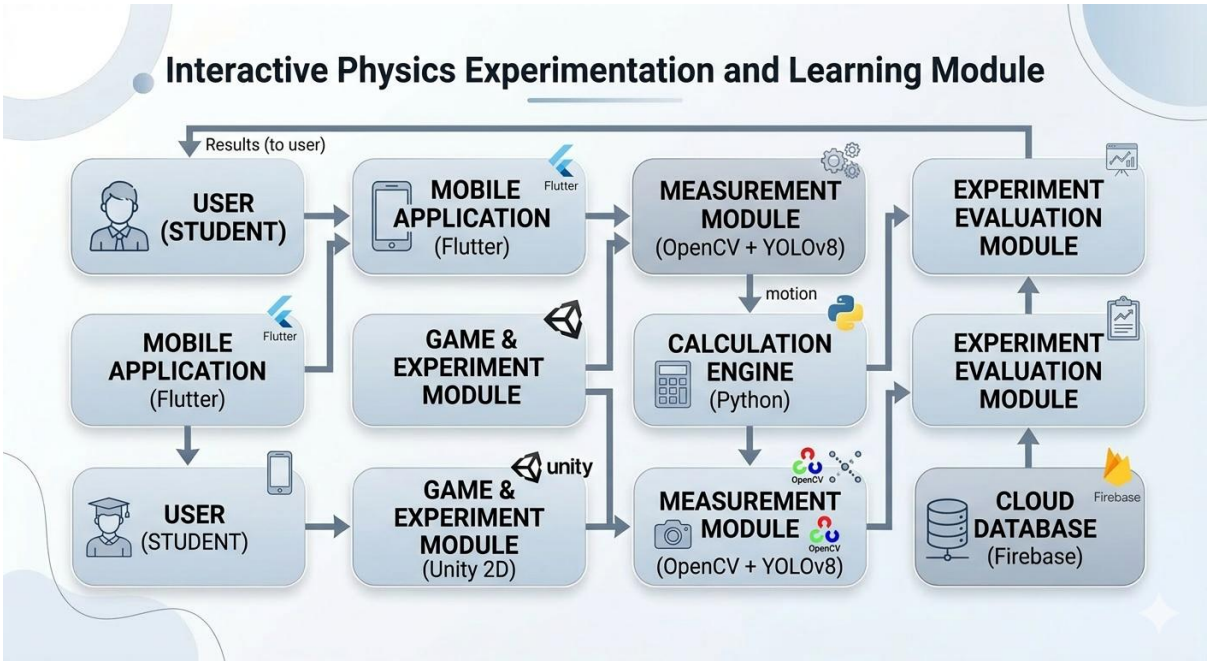


Figure 2 Research Methodology Process Diagram

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed solution is a mobile-based interactive physics learning system designed to support students in understanding physics concepts through games, measurements, calculations, and virtual experiments. The high-level architecture consists of several connected components that work together to provide a complete learning experience. The main components are the mobile application interface**,** game and experiment module**,** measurement extraction module**,** calculation engine**,** experiment evaluation module**,** backend services, and cloud database. The mobile application, developed using Flutter, acts as the main access point for students. Through this interface, users can select physics games, perform virtual experiments, view generated measurements, and receive feedback. The game and experiment modul**e**, developed using Unity 2D, contains interactive simulations where students perform actions that produce measurable outputs such as distance, time, and velocity. This is the core individual component of the system, and it consists of two parts: the physics game section and the virtual experiment section.

The measurement extraction module uses OpenCV and YOLOv8 to identify objects, track movement, and capture measurement-related values from the simulations. These values are passed to the calculation engine, implemented in Python, where physics formulas are applied to generate results and step-by-step explanations. In parallel, the experiment evaluation module analyzes student performance in virtual practical tasks based on apparatus selection, safety tools, procedural steps, and final result accuracy. All processed data, user progress, and results are managed by backend services and stored in Firebase for real-time access and synchronization.

The interaction between components follows a continuous flow. First, the user interacts with a game or experiment through the mobile app. The game environment sends simulation data to the measurement extraction module. After extracting the necessary values, the system forwards them to the calculation and evaluation modules. The generated outputs are then returned to the mobile app and displayed to the student as measurement results, calculation steps, and experiment feedback. This data flow ensures that student interaction is directly transformed into meaningful academic guidance.

From a scalability perspective, Firebase supports real-time cloud storage and multiple concurrent users, making the system suitable for large school deployments. In terms of security, user data is stored securely with controlled access, and only necessary learning-related information is collected to protect privacy and confidentiality.

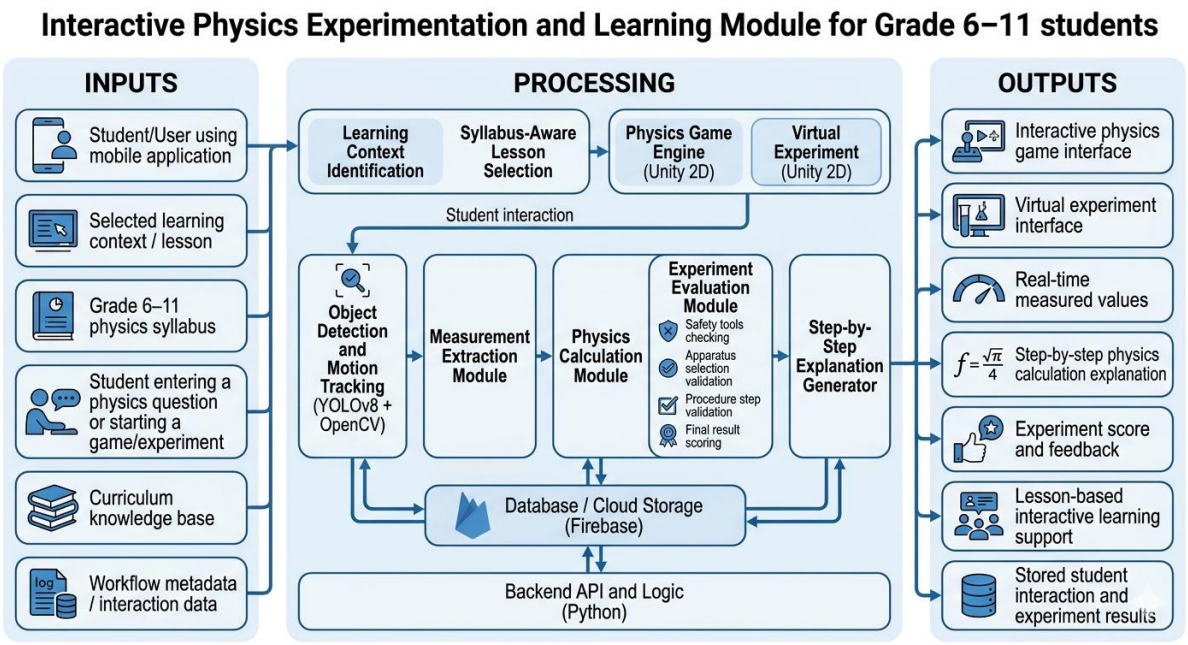


Figure 03 High Level System Architecture Diagram

5. USER REQUIREMENTS

- Stakeholders
  1. **Students**– Grade 6–11 students who use the mobile application to learn physics concepts through interactive games and virtual experiments.
  2. **School Administrators** – Responsible for supporting the adoption of the system in schools and encouraging its use for educational purposes.
  3. **Developers / Researchers** – Responsible for designing, developing, maintaining, and improving the learning module.
  4. **Educational Institutions** – Schools or educational organizations that may adopt the system to enhance digital learning and support physics education.

- Functional requirements
  - The system shall allow users to play interactive physics-based games.
  - The system shall capture real-time measurements such as distance, time, and velocity from user interactions.
  - The system shall extract measurement data using object detection and motion tracking techniques.
  - The system shall apply physics formulas automatically based on generated values.
  - The system shall display step-by-step calculation explanations to enhance understanding.
  - The system shall allow users to perform virtual laboratory experiments using digital tools.
  - The system shall evaluate experiment performance based on apparatus selection, safety, procedural steps, and accuracy of results.
  - The system shall store user data and progress in a cloud database.
  - The system shall provide feedback and results to users after each activity.
  - The system shall support mobile-based access through a user-friendly interface

- Non-functional requirements
  - **Performance:** The system should provide fast response times and real-time interaction without delays.
  - **Usability:** The interface should be simple, intuitive, and suitable for school students.
  - **Security:** User data must be securely stored, and unauthorized access should be prevented.
  - **Scalability:** The system should support multiple users simultaneously without performance degradation.
  - **Reliability:** The system should function consistently without crashes or data loss.
  - **Compatibility:** The application should run smoothly on different mobile devices and operating systems.
  - **Maintainability:** The system should be easy to update and extend with new features or experiments.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The primary target market for the proposed system includes school students (Grade 6–11), particularly in regions where access to physical laboratory facilities is limited. Secondary markets include schools, tuition centers, and educational institutions seeking digital learning solutions. The key customer persona is a student who struggles with understanding physics concepts due to lack of practical exposure, as well as teachers who require interactive tools to improve student engagement and learning outcomes. Additionally, education authorities and private learning platforms can adopt the system for wider implementation.

Value Proposition

The proposed system offers a unique value by combining game-based learning, real-time measurement generation, automated physics calculations, and virtual experiment evaluation in a single mobile application. Unlike existing solutions that focus on only one aspect (simulation, games, or labs), this system provides a comprehensive and interactive learning experience**.** It improves conceptual understanding, increases student engagement, and provides accessible practical learning without requiring physical laboratory infrastructure.

Revenue Model

The revenue model for this system can be based on a freemium educational platform approach. Basic features such as access to selected physics games and learning modules can be offered for free to encourage adoption by schools and students. Advanced features, including additional experiments, detailed analytics, and extended learning modules, can be provided through subscription-based plans for schools or institutions. Schools may subscribe to a yearly or monthly plan to access the full platform for their students. Additional revenue opportunities may also include institutional licensing, partnerships with educational organizations, and integration with digital learning programs**.**

Cost Estimation

The main costs associated with developing and maintaining the system include software development, system maintenance, cloud services, and content creation**.** Development costs involve building the mobile application using Flutter, developing backend services using Python, integrating Firebase for cloud database management, and creating interactive games and simulations using Unity 2D. Additional costs may include cloud hosting services, security implementation, testing, and continuous updates. Operational costs may also include technical support, server maintenance, and future development for adding new experiments or learning features.

Pricing Strategy

The pricing strategy should focus on making the platform affordable and accessible for educational institutions, particularly schools in developing regions. **A** tiered pricing model can be used where individual users have access to basic features for free, while schools can purchase institutional packages that allow multiple students and teachers to use the system. Pricing may include monthly or yearly subscription plans for schools, with discounts for larger institutions or government educational programs. This strategy ensures that the system remains accessible to a wide range of users while still supporting sustainable platform development.

Competitive Advantage

The proposed system has several competitive advantages compared to existing educational platforms. First, it integrates physics-based games, virtual experiments, real-time measurement generation, and step-by-step calculation explanations within a single learning environment**,** which many current platforms do not provide. Second, the mobile-based design makes the platform accessible and easy to use for students in rural areas where laboratory facilities may be limited. Third, the system focuses on transforming student interaction into measurable physics learning outcomes, allowing students to understand concepts through observation, experimentation, and calculation rather than memorization.

Intellectual Property Considerations

The developed system, including software design, algorithms, and user interface, can be protected under copyright laws. Any unique methods related to measurement extraction, calculation generation, or evaluation processes may be considered for intellectual property protection. Proper licensing will be ensured for third-party tools and libraries such as OpenCV and Firebase. This helps secure ownership and supports future commercialization opportunities.

7. BUDGET AND JUSTIFICATION

- **Hardware costs**
  The proposed research does not require any specialized hardware devices, as the system is developed as a mobile-based software application. Development and testing activities can be carried out using existing resources such as personal laptops and smartphones owned by the research team. Since the system focuses on virtual simulations and digital experiments rather than physical laboratory equipment, additional hardware purchases are not necessary. Therefore, the estimated hardware cost for this project is minimal or zero.

- **Software licenses**
  The project utilizes mainly open-source and free development tools. Flutter is used for mobile application development, Python is used for backend processing, and Firebase provides a free-tier cloud service suitable for small-scale applications. Unity 2D also offers a free personal license for educational purposes. OpenCV and YOLOv8 are open-source technologies used for computer vision and object detection. As a result, no significant cost is required for software licenses, making the project highly cost-effective.

- **Cloud services**
  The system uses Firebase as the cloud platform for data storage, authentication, and real-time database management. Firebase offers a free-tier plan that is sufficient for development, testing, and demonstration purposes in this research. This allows the system to operate efficiently without incurring additional cloud service costs during the research phase. Therefore, cloud service expenses are expected to be minimal or zero.

- **Human effort cost**
  The development, design, and implementation of the system are carried out by the research team as part of their academic research work. Since this project is conducted within an educational context, human effort is not calculated as a financial cost. Instead, it is considered as a contribution of time, knowledge, and skills by the students and supervisors involved in the project.

- **Justification for each expense**
  The proposed system is designed to be highly cost-effective by utilizing existing devices and open-source technologies. The absence of hardware requirements and the use of free development tools significantly reduce the overall budget. Firebase's free cloud services further minimize operational costs. This approach demonstrates that the system can be developed and implemented with minimal financial investment, making it practical and scalable for educational institutions, especially in resource-limited environments. Additionally, the use of widely available technologies ensures that the system can be easily maintained, extended, and adopted in the future without requiring significant financial resources.

Table 1 Budget and Budget Justification

Budget Category	**Item/Resource**	**Estimated Cost (LKR)**	**Justification**
Hardware Costs	Existing laptops and smartphones	0	Development and testing will use personal devices owned by the research team. No additional hardware is required.
Software Licenses	Flutter, Python, Unity 2D, OpenCV, YOLOv8	0	All tools are open-source or free for educational use
Cloud Services	Firebase Free Tier	0	Firebase provides free database, authentication, and cloud services sufficient for development and testing.
Development Tools	Visual Studio Code	0	These development environments are freely available for software development
Human Effort Cost	Student research team work		The development and implementation are part of the academic research work of the students.
Total Estimated Budget		0	

8. WORK BREAKDOWN STRUCTURE (WBS)

Table 2 Work Breakdown Structure

WBS ID	Task Name	Member	Description	Duration
1.0	Project Initiation	All	Identify research problems, define project scope, and obtain supervisor approval	Month 1
1.1	Literature Review	All	Review research papers related to virtual labs, mobile learning, learning analytics, adaptive feedback, RAG-based chatbots, gamification, and personalized learning systems	Month 1 – Month 2
1.2	Research Proposal Preparation	All	Prepare research proposal including introduction, research gaps, objectives, and methodology for all four modules	Month 2
2.0	System Requirement Analysis	All	Identify functional and non-functional requirements of the complete mobile virtual physics laboratory system	Month 3
2.1	System Architecture Design	All	Design overall system architecture including Flutter mobile application, Python backend services, Firebase, and database structure	Month 3
2.2	Database Design	All	Design Cloud Firestore data structure for user profiles, quiz scores, experiment data, analytics records, chatbot data, game records, learning paths, and curriculum content	Month 3
3.0	User Authentication and Profile Module	Wijerathna C D W	Implement student registration, login, Google Sign-In, guest access, and student profile management using Firebase Authentication and Cloud Firestore	Month 4

3.1	Intelligent Quiz Evaluation System	Wijerathna C D W	Develop intelligently generated sub-lesson quizzes with MCQs and structured questions aligned to Grade 6–11 curriculum, with shuffled retake support	Month 5 – Month 6
3.2	Experiment Performance Evaluation	Wijerathna C D W	Implement system to evaluate practical experiment steps, record performance outcomes, and identify poorly performed experiment areas per sub-lesson	Month 6 – Month 7
3.3	Learning Analytics and Weakness Detection	Wijerathna C D W	Develop algorithms to analyze sub-lesson quiz and experiment results and detect weak theory and practical areas	Month 7 – Month 8
3.4	Adaptive Feedback Generation	Wijerathna C D W	Implement rule-based feedback mechanism to generate personalized feedback messages and display improvement suggestions on the student home page	Month 8
3.5	Personalized Learning Path Assignment	Wijerathna C D W	Develop learning path assignment system that detects weak students and assigns targeted resources including explanations, guided examples, experiment procedures, and video materials; enable reattempt after completion; assign additional external resources for persistent weakness	Month 8 – Month 9
3.6	Dashboard and Visualization Development	Wijerathna C D W	Develop student dashboard with sub-lesson performance indicators, quiz and experiment scores, weak area summaries, feedback messages, and learning path access	Month 9 – Month 10
4.0	Grade-Tagged Physics Knowledge Base	Sachintha Nirmal S.M	Build a grade-tagged physics knowledge base containing curriculum-aligned physics content organized by grade level,	Month 4 – Month 5

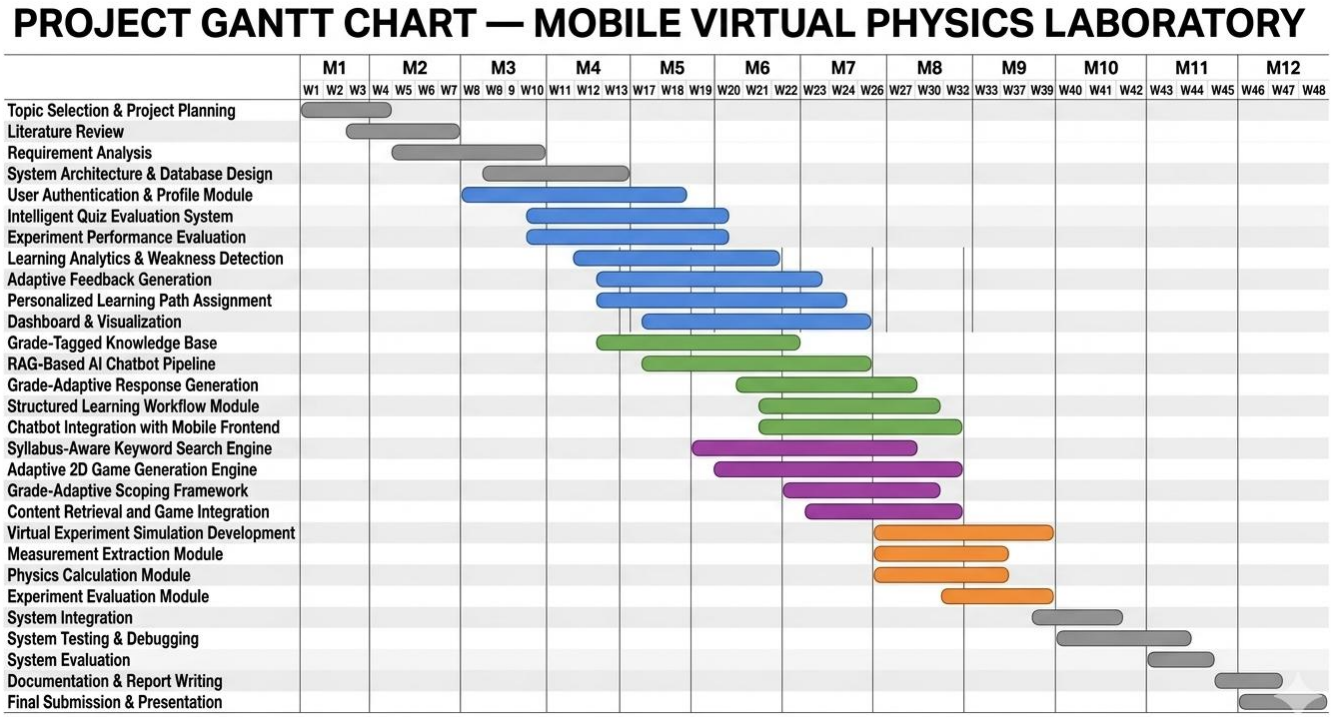
			lesson, topic, and activity for RAG retrieval	
4.1	RAG-Based AI Chatbot Pipeline	Sachintha Nirmal S.M	Implement the Retrieval-Augmented Generation pipeline using LLM to retrieve relevant syllabus content and generate physics explanations for student queries	Month 5 – Month 6
4.2	Grade-Adaptive Response Generation	Sachintha Nirmal S.M	Develop grade-adaptive response generation that tailors chatbot answers based on the student's grade level and lesson context	Month 6
4.3	Structured Learning Workflow Module	Sachintha Nirmal S.M	Design and implement the structured learning workflow including hierarchical navigation for grades, lessons, topics, experiments, and activities	Month 4 – Month 5
4.4	Chatbot Integration with Mobile Frontend	Sachintha Nirmal S.M	Integrate the AI chatbot interface and structured learning workflow screens into the Flutter mobile application	Month 7
5.0	Syllabus-Aware Keyword Search Engine	Arthanayake C T H	Implement keyword-based search engine that matches student-entered physics keywords with Sri Lankan National curriculum syllabus materials	Month 4 – Month 6
5.1	Adaptive 2D Game Generation Engine	Arthanayake C T H	Create curriculum-aligned 2D educational games using Unity that adapt game difficulty and content based on student grade level and lesson topic	Month 5 – Month 7
5.2	Grade-Adaptive Scoping Framework	Arthanayake C T H	Design scoping framework that maps digital learning activities and game difficulty to structured lesson-level objectives for Grades 6–11	Month 6 – Month 7

5.3	Content Retrieval and Game Integration	Arthanayake C T H	Integrate keyword-based retrieval system and game generation module with Firebase Firestore for NIE-aligned curriculum data storage and retrieval	Month 7 – Month 8
6.0	Virtual Experiment Simulation Development	Vidurangi H I D G	Develop interactive physics virtual experiments using Unity 2D where students can interact with virtual objects and perform guided experiments	Month 4 – Month 7
6.1	Measurement Extraction Module	Vidurangi H I D G	Implement real-time measurement extraction system using OpenCV and YOLOv8 to capture physics measurements such as distance, time, and velocity from student interactions	Month 5 – Month 6
6.2	Physics Calculation Module	Vidurangi H I D G	Apply physics formulas to extracted measurements to generate step-by-step calculation results and concept explanations for students	Month 6 – Month 7
6.3	Experiment Performance Evaluation Module	Vidurangi H I D G	Implement scoring system that evaluates student experiment performance based on safety tool usage, apparatus selection, procedure steps, and final result accuracy	Month 7 – Month 8
7.0	System Integration	All	Integrate Flutter mobile application, Python backend analytics services, Firebase Authentication, Cloud Firestore, chatbot, experiment module, gamified learning, and analytics module into a complete unified system	Month 10 – Month 11
8.0	System Testing and Debugging	All	Conduct functional and performance testing of all modules including authentication, quiz evaluation, experiment evaluation, weakness detection, feedback	Month 10 – Month 11

			generation, learning path assignment, chatbot accuracy, RAG retrieval, experiment simulations, and game functionality	
9.0	System Evaluation	All	Evaluate usability and effectiveness of the complete system including mastery level prediction accuracy, chatbot performance, RAG retrieval accuracy, experiment evaluation accuracy, search relevance, game engagement, and learning path effectiveness	Month 11
10.0	Documentation and Final Report	All	Prepare final research documentation including results, discussion, and complete system description for all four modules	Month 11 – Month 12
11.0	Final Presentation Preparation	All	Prepare presentation slides, demonstration materials, and final system submission	Month 12

9. GANTT CHART

Table 3 Gantt Chart



10. References List

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11.APPENDICES

APPENDIX 1 Survey instruments

This research does not include survey-based data collection methods. Instead, the system evaluation is conducted through user interaction testing and performance analysis. Therefore, no survey instruments are included in this study.

APPENDIX 2 Detailed diagrams

This section includes detailed diagrams related to the system design and methodology. These diagrams provide a visual representation of the research workflow and system structure. Key diagrams include:

- Research Methodology Diagram
- Research Methodology Process Diagram
- High-Level System Architecture Diagram

These diagrams illustrate the flow of activities, system components, and interactions between modules such as the mobile application, game engine, measurement module, calculation engine, and cloud database.

APPENDIX 3 Preliminary UI mockups

Preliminary user interface (UI) mockups are included to demonstrate the design of the mobile application. These mockups present the layout and structure of key screens such as:

- Home screen (navigation to games and experiments)
- Physics game interface (interaction and measurement display)
- Calculation screen (step-by-step explanation of formulas)
- Virtual laboratory interface (experiment setup and tools)
- Result and feedback screen

These mockups help visualize how users will interact with the system and ensure a user-friendly design suitable for school students.

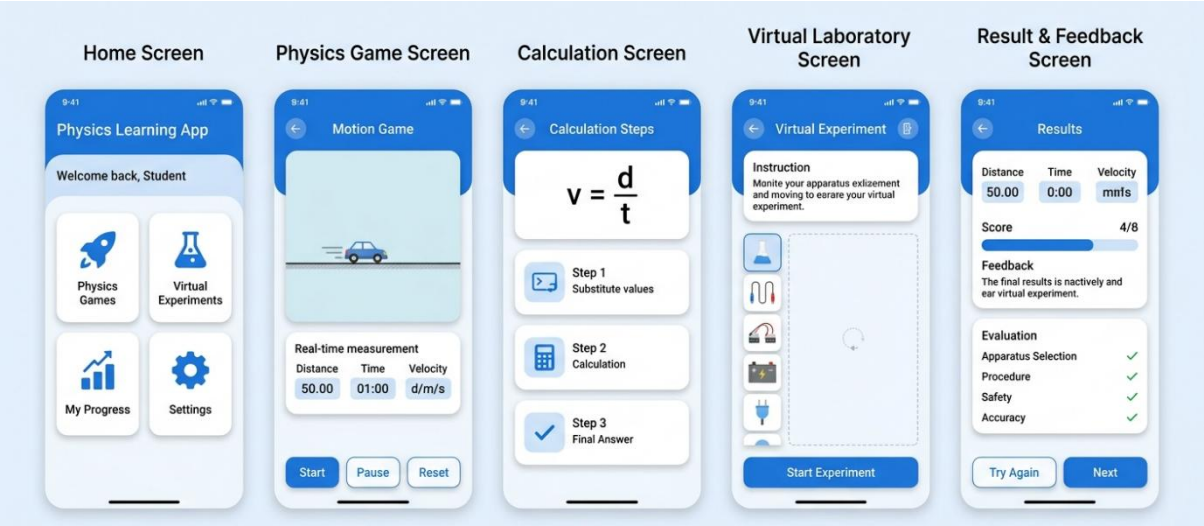


Figure 04 Preliminary UI mockups

APPENDIX 4 Risk analysis

Table 4 Risk Analysis

Risk Type	Description	Mitigation Strategy
Technical Risk	Challenges in accurate measurement extraction using computer vision	Use optimized models(YOLOv8) and test with multiple scenarios
Performance Risk	Possible delays in real-time processing	Optimize algorithms and use efficient backend processing
User Adoption Risk	Students or teachers may be unfamiliar with the system	Provide simple UI design and user guidance
Data Security Risk	Risk of unauthorized access to user data	Use secure Firebase authentication and data protection mechanisms
System Reliability Risk	Application crashes or unexpected bugs	Perform thorough testing and continuous updates

APPENDIX 5 Ethical documents

Ethical considerations are maintained throughout the research and system implementation. The following principles are followed:

- User data is collected only for educational and research purposes.
- No sensitive personal information is stored in the system.
- User privacy and confidentiality are ensured through secure data handling.
- Participation in system testing is voluntary, and consent is obtained where required.
- The system is designed to support fair and inclusive access to education without bias